

Structuring Health Monitoring Dataset

Our solution uses a conventional machine-learning baseline, by compressing each stress time series into common features like mean, standard deviation, maximum, minimum, and root-mean-square. We then used these features to train a Random Forest regression model. Using feature engineering, we realised that adding stress-range and time-difference features improved the Random Forest score.

We applied rainflow counting to each dynamic-stress test time-series data to identify individual stress cycles and their corresponding stress ranges and cycle counts. Using the given Miner's Linear Cumulative Damage Rule, the Stress-life Curve, and the Rainflow Counting Method, we developed a fatigue indicator by summing the contribution of the rainflow counted cycles. For each cycle range, its number of occurrences is multiplied by a stress range raised to a particular power. We raise stress to a power as large stress cycles contribute disproportionately more to fatigue. Initially, we tested exponents of 3, 4 and 5, and found out that the correlation between the fatigue indicator and the known cumulative damage labels increased. By using 5-fold cross-validation for values close to 5, we found that an exponent of approximately 5.03 will produce the best validation result.

Our final model takes the rainflow-counted cycles from each signal, calculates fatigue indicator using the exponent 5.03, then multiplies this indicator by a proportionality constant learned from labelled training data to obtain predicted cumulative damage value.

Our solution was implemented using Python. The loading and processing of the large CSV time-series data were done on Pandas, while numerical calculations were done using NumPy. Skikit-learn was used to develop and evaluate our Random Forest model, perform 5-fold cross-validation and calculate MAPE were done using scikit-learn. We also utilised the Python rainflow package to identify individual stress cycles within each dynamic stress-time series and obtain their stress ranges and cycle counts.